

Oracle GI Patch Script v2

Configuration File (**patching.conf**)

```
GRID_HOME=/u01/app/19.0.0/grid
OPATCH_HOME=/u01/app/19.0.0/grid/OPatch
PATCH_ZIP=/u01/software/patches/p37145387_190000_Linux-x86-64.zip
LOG_DIR=/u01/app/patch_logs

# File ownership (usually Grid Infrastructure owner)
INSTALL_USER=grid
INSTALL_GROUP=oinstall

# yes|no
DRY_RUN=yes
```

Script (**patch_grid.sh**)

```
#!/bin/bash
#
=====
=====
# Script Name: patch_grid.sh
# Description: Automate GI Patching with opatchauto and individual sub-
patch checks.
# MUST BE RUN AS ROOT
#
=====
=====

set -euo pipefail

if [[ $EUID -ne 0 ]]; then
    echo "ERROR: This script must be run as ROOT (since it uses
opatchauto)."
    exit 1
fi
```

```

CONFIG_FILE="${1:-./patching.conf}"

if [[ ! -f "$CONFIG_FILE" ]]; then
    echo "ERROR: Config file not found: $CONFIG_FILE"
    exit 1
fi

source "$CONFIG_FILE"

: "${GRID_HOME:?GRID_HOME is required in config}"
: "${OPATCH_HOME:?OPATCH_HOME is required in config}"
: "${PATCH_ZIP:?PATCH_ZIP is required in config}"
: "${INSTALL_USER:=grid}"
: "${INSTALL_GROUP:=oinstall}"

LOG_DIR="${LOG_DIR:-$PWD/logs}"
DRY_RUN="${DRY_RUN:-no}"

shopt -s nocasematch
if [[ "$DRY_RUN" =~ ^(yes|true|1)$ ]]; then DRY_RUN="yes"; else
DRY_RUN="no"; fi
shopt -u nocasematch

mkdir -p "$LOG_DIR"
chown "$INSTALL_USER":"$INSTALL_GROUP" "$LOG_DIR"

TS=$(date +%Y%m%d_%H%M%S)
ANALYZE_LOG="$LOG_DIR/opatchauto_analyze_$TS.log"
PATCH_LOG="$LOG_DIR/opatchauto_patch_$TS.log"
UNZIP_DIR="$LOG_DIR/patch_unzip_$TS"

export ORACLE_HOME="$GRID_HOME"
export PATH="$OPATCH_HOME:$GRID_HOME/bin:$PATH"

run_cmd() {
    local cmd="$1"
    if [[ "$DRY_RUN" == "yes" ]]; then
        echo "[DRY-RUN] $cmd"
        return 0
    fi
    echo "[EXECUTING] $cmd"
    eval "$cmd"
}

echo "-----"
echo "  Oracle GI Patching Automator (19c)"
echo "-----"

```

```

echo "GRID_HOME      : $GRID_HOME"
echo "INSTALL_USER   : $INSTALL_USER"
echo "PATCH_ZIP      : $PATCH_ZIP"
echo "DRY_RUN         : $DRY_RUN"
echo "-----"

chown "$INSTALL_USER":"$INSTALL_GROUP" "$0"
chmod 750 "$0"

[[ ! -d "$GRID_HOME" ]] && { echo "ERROR: GRID_HOME not found"; exit
1; }
[[ ! -f "$PATCH_ZIP" ]] && { echo "ERROR: PATCH_ZIP not found"; exit
1; }

echo ">>> Checking OPatch version..."
run_cmd "\"$OPATCH_HOME/patch\" version | tee \"$LOG_DIR/
opatch_version_$TS.log\""

echo ""
echo ">>> Running opatchauto ANALYZE..."
run_cmd "\"$GRID_HOME/OPatch/opatchauto\" apply \"$PATCH_ZIP\" -analyze
| tee \"$ANALYZE_LOG\""

echo ""
echo ">>> Preparing Unzip Directory..."
run_cmd "rm -rf \"$UNZIP_DIR\" && mkdir -p \"$UNZIP_DIR\""
run_cmd "unzip -q \"$PATCH_ZIP\" -d \"$UNZIP_DIR\""
run_cmd "chown -R $INSTALL_USER:$INSTALL_GROUP \"$UNZIP_DIR\""

echo ""
echo ">>> Inspecting Sub-folders and running 4 Pre-reqs..."
if [[ "$DRY_RUN" == "no" ]]; then
    PATCH_DIRS=$(find "$UNZIP_DIR" -mindepth 1 -maxdepth 2 -type d)
else
    echo "[DRY-RUN] Would search subfolders in $UNZIP_DIR"
    PATCH_DIRS=()
fi

for pdir in "${PATCH_DIRS[@]}; do
    if [[ ! -f "$pdir/etc/config/actions.xml" && ! -f "$pdir/etc/config/
inventory.xml" ]]; then
        continue
    fi

    base=$(basename "$pdir")
    echo ""
    echo "--- Sub-patch: $base ---"

```

```
run_cmd "su - $INSTALL_USER -c \"cd $pdir && $OPATCH_HOME/patch
prereq CheckConflictAgainstOHWithDetail -ph .\" | tee \"\$LOG_DIR/
prereq_${base}_01_conflict_$TS.log\""
run_cmd "su - $INSTALL_USER -c \"cd $pdir && $OPATCH_HOME/patch
prereq CheckIfPatchApplied -ph .\" | tee \"\$LOG_DIR/prereq_${base}
_02_applied_$TS.log\""
run_cmd "su - $INSTALL_USER -c \"cd $pdir && $OPATCH_HOME/patch
prereq CheckApplicable -ph .\" | tee \"\$LOG_DIR/prereq_${base}
_03_applicable_$TS.log\""
run_cmd "su - $INSTALL_USER -c \"cd $pdir && $OPATCH_HOME/patch
prereq CheckSystemSpace -ph .\" | tee \"\$LOG_DIR/prereq_${base}
_04_space_$TS.log\""
done

echo ""
echo ">>> Final Step: Running opatchauto APPLY..."
run_cmd "\"\$GRID_HOME/OPatch/opatchauto\" apply \"\$PATCH_ZIP\" | tee
\"\$PATCH_LOG\""

echo ""
echo "Workflow Finished. Logs available in: $LOG_DIR"
```